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SiC leads to both lower system cost and an improvement of efficiency

Did you know that silicon carbide (SiC) could improve efficiency of power products whilst simultaneously lower the total product cost? Niranjan G., general manager, Business Development and Technical Marketing, ROHM Semiconductor speaks to Dilin Anand and Sneha Ambastha from EFY at an industry trade show in Delhi

Most important question first: What is the biggest takeaway for an engineer using silicon carbide (SiC) based solutions?

Let us take the example of a solar inverter to consider silicon carbide's (SiC's) contribution. Solar cells are a very non-efficient way of converting energy since their efficiency is barely 40-50 per cent. Here, if we were using silicon (Si) for the inverter, we would lose much of that 40 per cent as well. This is where SiC comes in.

If the inverter's efficiency is 97 per cent, after using SiC, it would be 70-80 per cent. Thus, instead of using a bigger solar cell, we can use a smaller one with SiC without compromising its efficiency. Therefore SiC leads to a lower system cost along with improved efficiency.

What is it that enables SiC to be so much better than Si?

SiC is an attractive material from which we can manufacture power devices that will be comparatively better than those made from Si. This is because SiC has ten times the dielectric breakdown field strength and three times the band gap as well as thermal conductivity. This lets SiC devices withstand higher breakdown voltage, have lower resistivity and operate at higher temperature.

How does SiC look from the cost perspective?

The SiC technology is a very broad one. Although its volume-to-cost ratio is still not up to the point, it is similar to what Si technology was 50 years ago when vacuum tubes were being used. Vacuum tubes were very cheap and Si was very expensive. But now that SiC is in use, as its demand increases it would almost compete with Si. The transition to 100mm SiC wafers a few years back was a significant milestone towards reduced cost, and the next move to 150mm wafers will help in further cost reduction.

Where do design engineers come in the SiC story?

The initial traction for this technology is with the designers because if you want to reduce the price of a solar

cell, the first thing to do would be innovation. The second would be to reduce the panel size. Now, the question here is how to reduce the panel size because if we have 10kW of power, after reducing the panel size it would become, say, 6kW. Hence the only area where this reduction in size can take place is the inverter portion. The inverter is of Si which costs, say, \$100, which if I change to SiC inverter with an additional cost of \$100, it can help me save \$200 in the solar cell. This shows that the total cost comes down.

The initial traction for the designers was only for areas such as solar, wind, inverters and trains for motor drives; now companies have started using SiC even in air-conditioners. Slowly, even the consumer electronics sector has started using SiC. However, in the general market, people get scared by seeing the price, although it can be brought down depending on how good the total system size is.

Could you elaborate on the sensors targeting the lighting space?

There are two types of wireless sensors for lighting; one is called the human-body sensor and the other is radio-frequency (RF) sensor.

When you enter a room, the lights in that room turn on sensing your presence due to the human-body sensors. With the RF sensors you can control the lighting in your house through an iPhone or any other Wi-Fi device using an app. The sensors with RF module contribute to home-energy and building-energy management systems.

Are these lighting devices based on the 'Internet of things' paradigm?

Yes, we have two such application-based devices. One is RF-based with a range of 300m. But once we get it connected to Wi-Fi, we can operate it from anywhere in the world. So, we offer this technology with respect to lighting as an application. ●